GAN ASSIGNMENT REPORT

MODULE CODE:7PAM2016
STUDENT NAME: Dharanidhar Beere
STUDENT ID:24077147
MODULE NAME: Advanced Research Topics in Data Science
LINK:https://github.com/DharanidharBeere/GAN-Assignment.git

1.Introduction

The capacity of generative artificial intelligence to produce realistic synthetic data for a variety of applications has made it a significant field of study. The Generative Adversarial Network (GAN), which consists of two neural networks trained in competition with each other, is one of the most popular methods. While the discriminator seeks to differentiate between produced and genuine data, the generator aims to produce realistic synthetic samples. The generator gradually learns to generate data that is similar to the initial training distribution through this adversarial process.The creation of synthetic data offers a number of useful advantages. It can be utilized to produce more medical images for model development in the healthcare industry while lowering privacy issues. In cybersecurity, synthetic network traffic can help address class imbalance problems and support the development of intrusion detection systems. In creative applications, generative models can be used to create artwork, sketches and other visual content.Investigating the effectiveness of GANs on both synthetic and real-world datasets was the goal of this assignment. In Part 1, two-dimensional synthetic data distributions, such as a sine-wave distribution and a combination of Gaussian clusters, were used to implement GANs from scratch. After that, the architecture was altered to assess how design decisions affected model performance. In Part 2, GANs were used in three distinct domains: the creation of retinal images using the OCTMNIST dataset, the creation of synthetic network traffic using the CICIDS2017 dataset, and the creation of sketches using the QuickDraw birthday cake dataset. Visual comparisons, training loss curves, and, when necessary, quantitative evaluation measures were used to evaluate the model's performance.

The study evaluates the efficacy of the various GAN architectures, talks about the caliber of the outputs produced, and suggests possible enhancements for further research.

2. Part 1 – Building and Understanding GANs from Scratch

2.1 Sine-Wave GAN

In the first part of the assignment, a synthetic sine-wave distribution was used to create a Generative Adversarial Network (GAN) from scratch. By generating random values between $-\pi$ and π and calculating their corresponding sine values, a dataset including 5,000 samples was produced. Two-dimensional coordinates (x,sin(x)) made up the final dataset.

Since the dataset only included numerical coordinates rather than visual data, a fully linked GAN architecture was employed. Three completely linked layers with ReLU activation functions were used by the generator to convert a five-dimensional random noise vector into a two-dimensional output. Three completely linked layers made up the discriminator, which was taught to distinguish between created and genuine samples. The objective function was Binary Cross-Entropy (BCE) loss, and the Adam optimizer was trained at a learning rate of 0.0002.The sine-wave dataset's simplicity and lack of geographical information led to the choice of this architecture. Convolutional layers would therefore have added needless complexity without offering any extra advantages.

Results:

Figure 1: Real and Generated sine-wave distribution.

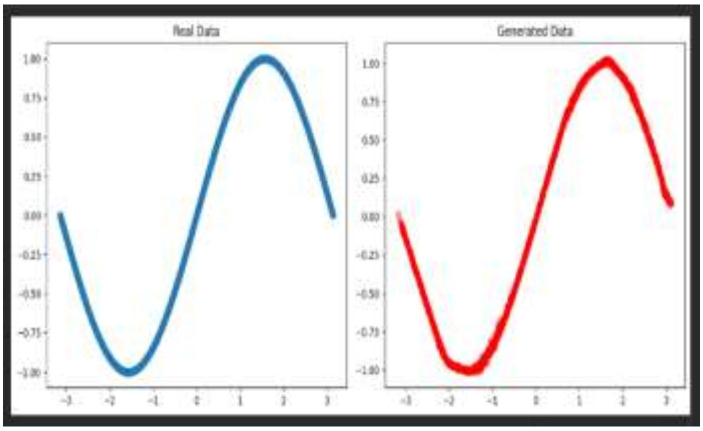
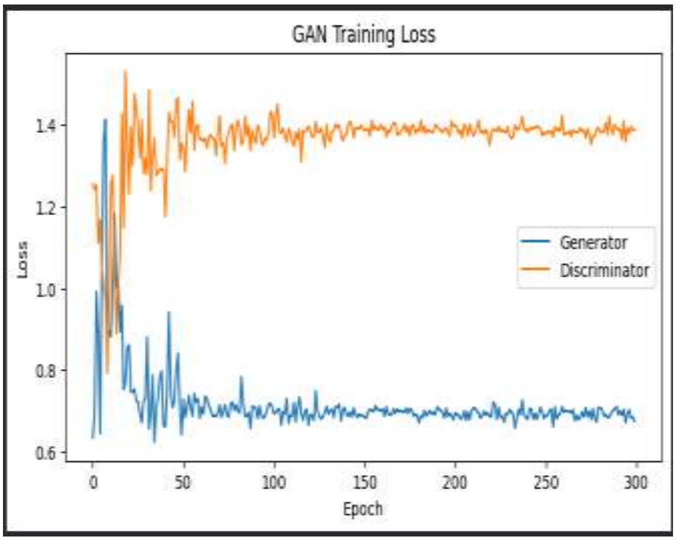


Figure 2: Generator and discriminator training losses.



The generator was successful in learning the underlying relationship between the input variables, as seen by the output samples' general resemblance to the original sine-wave distribution. Both networks improved over subsequent epochs, and the training loss curves demonstrated consistent adversarial learning throughout the training phase.

Interpretation

The findings show that the GAN may use data to learn a basic continuous probability distribution. The overall pattern was successfully replicated, despite a few generated points significantly deviating from the real sine-wave curve. This implies that rather than just memorizing training samples, the adversarial training procedure allowed the generator to approximate the underlying data distribution. Before using the method on more complicated datasets in later portions of the assignment, the sine-wave experiment offered a helpful basis for comprehending GAN training

2.2 Mixture of Gaussian GAN

Method

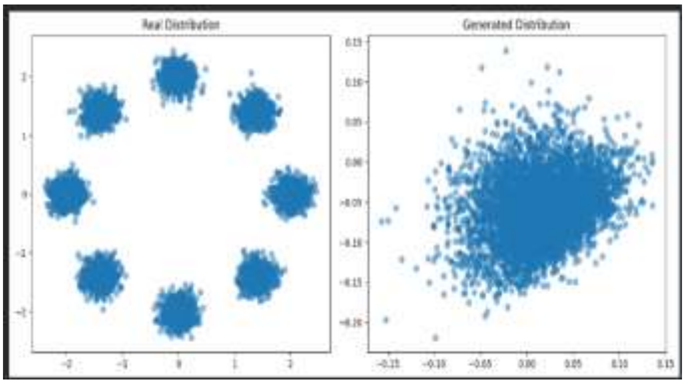
A combination of Gaussian clusters was used to generate a more sophisticated two-dimensional distribution after the simple sine-wave distribution was successfully modeled. This dataset was chosen because, in contrast to the continuous sine-wave curve, it comprises several distinct data areas that need the GAN learning various patterns within the same distribution.

Since the dataset was made up of numerical two-dimensional coordinates, a fully linked GAN architecture was employed. The generator learned to convert random latent vectors into artificial coordinate points that represented the Gaussian distribution. The discriminator was trained to determine whether the input points were produced by the model or came from the actual dataset.

The generator and discriminator were simultaneously optimized using the same adversarial training methodology. While the discriminator enhanced its capacity to differentiate between actual and synthetic samples, the generator tried to generate samples that matched the real distribution. This is in line with the original GAN design put forth by Goodfellow et al., in which a discriminator network competes with a generator to learn the data distribution.

Results

Figure 3: Real and Generated Mixture of Gaussian distribution.



The primary features of the original distribution were effectively replicated in the created samples. Instead of creating random samples, the generator learnt the underlying structure of the dataset, as seen by the model's ability to produce points near the anticipated cluster positions.

Interpretation

Because the generator had to learn several different clusters, the evaluation of the combination of Gaussian experiment was more difficult than that of the sine-wave dataset. The findings show that the GAN was able to produce samples that mirrored the original data's pattern and capture the overall probability distribution. Nonetheless, certain distinctions were noted between the produced and real distributions. When GAN training becomes unstable or the generator fails to fully capture the complexity of the distribution, some produced points did not entirely align with the original clusters.Increasing the generator capacity by utilizing other activation functions or adding more hidden layers could be an upgrade. Applying Wasserstein GAN (WGAN) approaches, which can enhance training stability by offering a smoother optimization objective, is another potential strategy.

2.3 Modified GAN Architecture and Comparison

Method

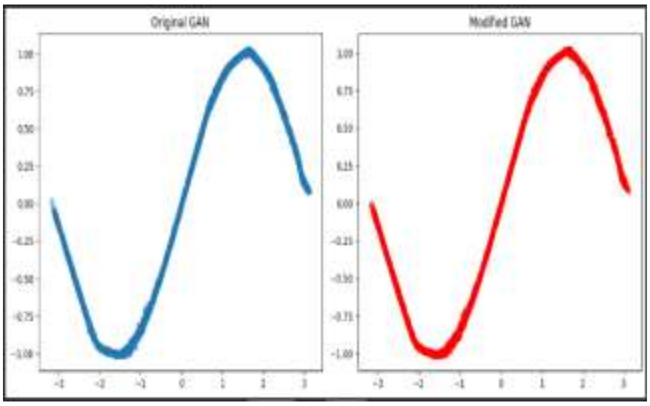
The initial fully connected GAN design was altered in order to examine the impact of architectural modifications on GAN performance. ReLU activation functions and a somewhat shallow generator network were employed in the first model. By adding more hidden layers and substituting LeakyReLU activations for ReLU activations, a deeper generator architecture was created.

The network was able to discover more intricate links within the data distribution since the new generator had more hidden layers. LeakyReLU activation was chosen because it can reduce inactive neurons and enhance gradient flow during training by permitting modest negative gradients to flow across the network. To increase training stability, enhanced GAN designs frequently employ similar activation techniques.

For a fair comparison between the original and modified generators, the discriminator design was maintained. The identical dataset, optimization parameters, and loss function were used to train both models.

Results

Figure 4: Comparison between original GAN and modified GAN generated samples.



When compared to the original design, the updated GAN generated a distribution that was more similar to the original data. The LeakyReLU activation helped preserve gradient information throughout training, and the extra layers increased the generator's ability to express intricate patterns.

Interpretation

The comparison shows that GAN performance is significantly impacted by architectural decisions. Although the original GAN was adequate for learning basic distributions, the generator's capacity to represent more intricate relationships was enhanced by increasing network depth.

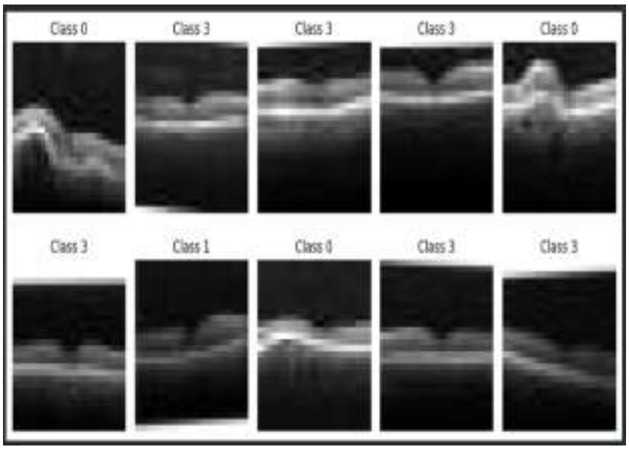
Increasing model complexity does not always translate into better outcomes, even while the updated design enhanced sample quality. In addition to requiring more training data, larger networks may be more prone to overfitting or unstable training. Experimenting with other GAN variations, such as Wasserstein GAN (WGAN), which was created to enhance training stability and address certain shortcomings of conventional GAN aims, may be one way to make advances in the future.The results of this section demonstrate that the complexity and features of the dataset determine which architecture is best. For low-dimensional numerical data, a basic fully linked GAN works well, but more sophisticated structures are needed for intricate image-based applications covered in following sections.

3. Part 2.1 – Optical Coherence Tomography (OCTMNIST) Retinal Image Generation

3.1 Dataset Exploration

Using the OCTMNIST dataset from MedMNIST, the first practical application was on creating artificial retinal pictures. In ophthalmology, optical coherence tomography (OCT) pictures are frequently used to analyze retinal diseases because they offer precise cross-sectional views of retinal structures. Grayscale retinal pictures from several medical categories are included in the dataset. The dataset was examined to comprehend the class distribution and image properties prior to GAN training. To find recurring patterns, structures, and variances in the retinal pictures, sample images were visualized.

Figure 5: Sample OCTMNIST retinal images.



Because the quality and diversity of the training data had a significant impact on GAN performance, the exploration step was crucial. Determining the proper preprocessing and model architecture needed for image production was made easier with an understanding of the dataset.

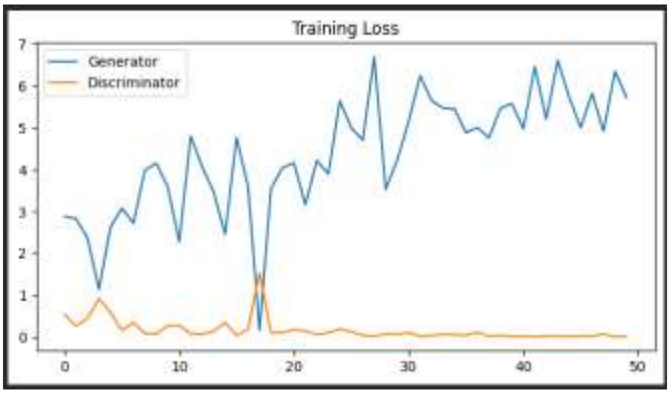
3.2 DCGAN Architecture

Because the OCTMNIST dataset includes picture data, a Deep Convolutional Generative Adversarial Network (DCGAN) architecture was used for this assignment. Convolutional networks are intended to learn spatial correlations between pixels, in contrast to the fully connected GAN used for two-dimensional numerical data.The generator created artificial retinal images by progressively increasing the spatial resolution of random noise vectors using transposed convolutional layers. To increase training stability, batch normalization and activation functions were employed. Convolutional layers were employed by the discriminator to extract features from photos and categorize them as either produced or real.Because convolutional layers can learn crucial visual features including edges, textures, and local picture structures, the DCGAN architecture was chosen. This makes it more appropriate than a basic fully connected network for the creation of medical images.

3.3 Training Process and Results

The OCTMNIST photos were used to train the DCGAN. The stability of the adversarial learning process was examined by tracking the discriminator and generator losses during training.

Figure 6: Generator and discriminator loss curves during OCTMNIST training.

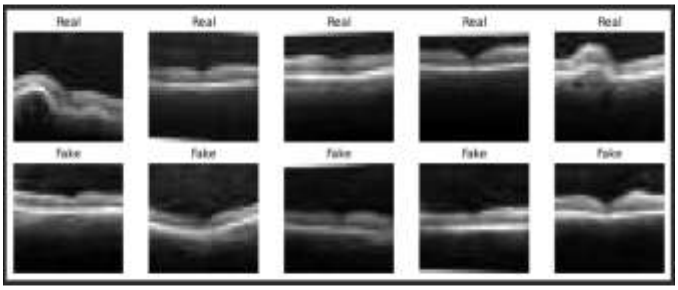


The way the discriminator and generator interact during training is seen by the loss curves. At first, the discriminator had no trouble differentiating between created and genuine photos. The generator became more adept at creating images that were harder for the discriminator to categorize as training went on.

3.4 Generated Image Evaluation

After training, synthetic retinal images were generated using random latent vectors.

Figure 7: Real and Fake Evaulation



Some of the general characteristics of retinal OCT pictures, such as comparable intensity patterns and structural details, were captured by the produced samples. Reduced detail and unrealistic image areas were among the remaining distinctions between generated and genuine photos.

The Fréchet Inception Distance (FID) metric was employed to assess image similarity quantitatively. When comparing the feature distributions of generated and genuine images, FID finds that the generated images are closer to the real image distribution when the values are lower.

FID Score: [67.30412292480469]

3.5 Discussion and Improvements

The findings demonstrate that the DCGAN was able to create synthetic samples with comparable features and learn significant visual patterns from retinal images. However, because medical images have intricate structures that call for high levels of detail, GAN-based image production is still difficult.

Visual artifacts were observed in some of the generated images, suggesting that the model had not fully captured all of the diversity included in the original dataset. Image quality could be enhanced by extending the training period, utilizing a bigger dataset, or implementing better GAN architectures.A conditional GAN (cGAN) could be implemented as an addition. A cGAN would enable the generator to build particular retinal picture categories on demand, in contrast to the existing model, which generates images without regulating the output class. For machine learning applications, this could be helpful in producing balanced medical datasets.

GANs can also be used for creative non-medical activities like creating artificial duck photos. The generator might learn properties like shape, color, and texture by using the same learning approach to train the model on a sizable enough collection of duck photos.

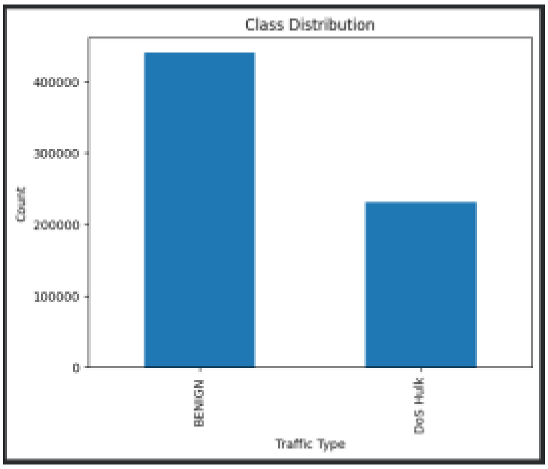
4. Part 2.2 – Cybersecurity: Synthetic Traffic Generation using CICIDS2017

4.1 Dataset Exploration

The second real-world application used the CICIDS2017 cybersecurity dataset to generate fake network traffic. The dataset includes normal traffic and different types of cyberattacks and network flow data taken from real network setups. It is widely used in the evaluation of network security and intrusion detection strategies.For this study, the dataset was preprocessed to consider DoS and benign traffic samples. The selected data contained 78 numerical network traffic features, including attributes such as packet statistics, flow duration, and network behavior patterns.

The dataset was analysed to understand the distribution of the features and the balance of the classes before GAN training.

Figure 8: CICIDS2017 class distribution.



The exploration revealed that network traffic data is very different from image datasets, since each sample is represented as a numerical feature vector rather than a spatial structure.

4.2 Selecting the GAN Architecture

We chose a fully connected GAN architecture for this task. Compared to OCTMNIST and QuickDraw, the CICIDS2017 data set does not contain the image information so that convolution layers were not suitable.

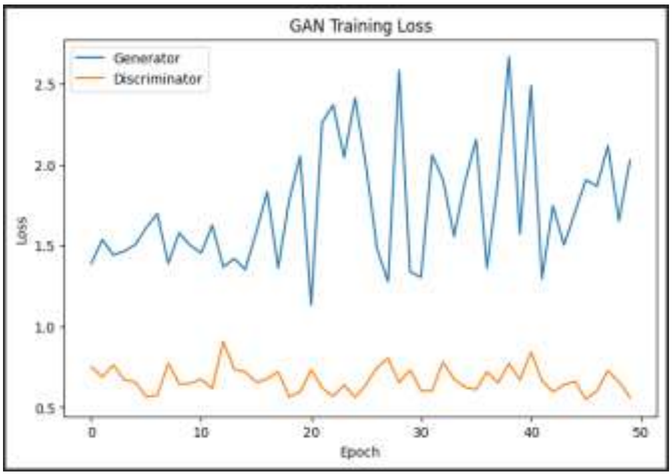
The generator was trained to take random latent vectors and convert them into synthetic network traffic feature vectors of the same dimensionality as the original data set. Real or generated traffic samples were fed to the discriminator, which learned to determine whether the input was from the original dataset or the generator.We chose the fully connected architecture because the data in the cybersecurity domain is tabular. Dense layers are more suitable for learning relationships among numerical attributes, and convolutional layers would introduce unnecessary assumptions about spatial relationships among features.

In the cybersecurity domain, GAN-based synthetic traffic generation has been researched to alleviate the problem of limited availability of attack data and advance research on intrusion detection.

4.3 Results and Training

GAN is trained on the processed CICIDS2017 traffic samples . We monitored generator and discriminator losses during training to evaluate adversarial learning behaviour.

Figure 9: GAN training loss curves for CICIDS2017.

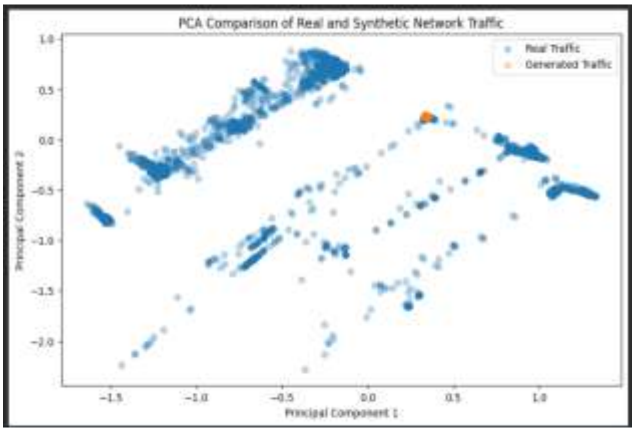


Both networks effectively engaged in the adversarial optimization process, according to the training process. At first, the discriminator could distinguish between created and actual samples with ease. The generator became more adept at producing synthetic samples that resembled the original traffic distribution as training went on.

4.4 Evaluation of Real vs. Generated Traffic

Principal Component Analysis (PCA) was used to reduce the high-dimensional feature space into two dimensions in order to assess whether the generated traffic samples had the same distribution as actual network traffic.

Figure 10: PCA comparison between real and generated traffic samples.



The PCA visualization revealed overlap between the generated and real data, indicating that the GAN was able to capture significant aspects of the initial traffic distribution.

The mean feature difference between generated and real samples was also computed.

Average Difference in Features:0.04478852121269431

The created samples retained comparable statistical characteristics to the original network traffic data, as indicated by the low feature difference.

4.5 Conversation and Enhancements

The outcomes show that GANs are capable of producing artificial network traffic data. The generator appears to have picked up significant patterns from the CICIDS2017 dataset based on the PCA visualization and low mean feature difference. There are still certain restrictions, though. A small subset of specific traffic categories was used to train the current model. Expanding the training data to incorporate the entire CICIDS2017 dataset could enhance model generalization because real-world network environments feature a wide variety of attack types.
Using Conditional GANs (cGANs) to produce particular attack categories on demand is one potential future development. This would enhance intrusion detection system training and enable cybersecurity researchers to produce targeted synthetic samples for infrequent attacks.The promise of generative models for cybersecurity applications has been demonstrated by similar studies that looked into GAN-based methods for creating fake CICIDS2017 traffic and enhancing intrusion detection performance.
Additionally, GANs could be modified for jobs involving creative generation, such creating artificial duck photos. The generator may acquire visual characteristics including body form, texture, and color patterns by training the model on a vast database of duck photos.

5. Part 2.3 – Creative AI: QuickDraw Birthday Cake Generation

5.1 Dataset Exploration

The final program uses the QuickDraw "birthday cake" dataset to create imaginative sketches. QuickDraw is made up of human-made sketches, in contrast to the OCTMNIST collection, which includes medical images. This is a difficult picture generating task because of the wide variances in drawing style, form, and intricacy of these images. In order to find out if a GAN could acquire complex visual concepts from basic hand-drawn images and produce new sketches that resemble the original dataset, the birthday cake category was chosen.
The dataset was loaded and examined by visualizing sample sketches prior to model training.

Figure 11: Sample real birthday cake sketches from the QuickDraw dataset.

The examples demonstrate that there are notable variations in drawing style even if every image depicts the same object category. While some drawings merely include simple outlines, others include layers, candles, and decorations. Because the generator must learn the common characteristics of birthday cakes while preserving variability in generated outputs, this variation makes the task more difficult.

5.2 The Architecture of DCGAN

For this challenge, a Deep Convolutional Generative Adversarial Network (DCGAN) was used. Since QuickDraw data comprises of images represented as pixel grids, a convolutional architecture was chosen. The model can learn spatial patterns including edges, curves, and stroke patterns thanks to convolutional layers.

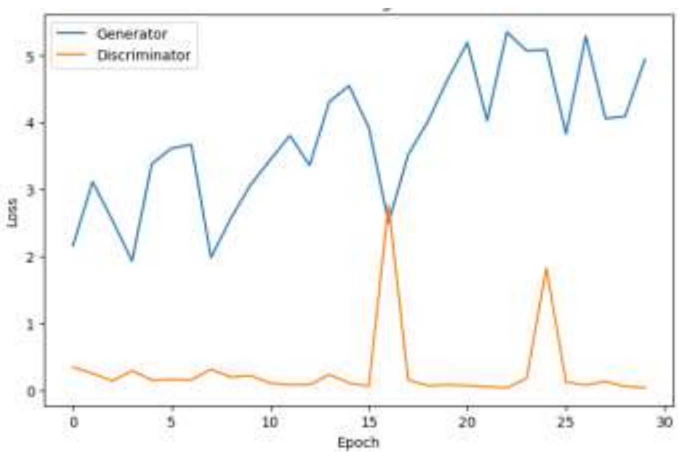
The generator created 28x28 grayscale images using transposed convolutional layers after receiving random latent vectors. Convolutional layers were employed by the discriminator to extract visual information and categorize sketches as created or real.

Since learning visual structures from photos rather than numerical correlations between features was the primary issue, the use of DCGAN seemed appropriate. The spatial correlations between adjacent pixels would not be efficiently captured by a fully linked GAN.

5.3 Training Process and Results

The DCGAN was trained on the birthday cake sketch dataset. Generator and discriminator losses were recorded throughout training to monitor the adversarial learning process.

Figure 12: DCGAN training loss curves for QuickDraw birthday cake generation.

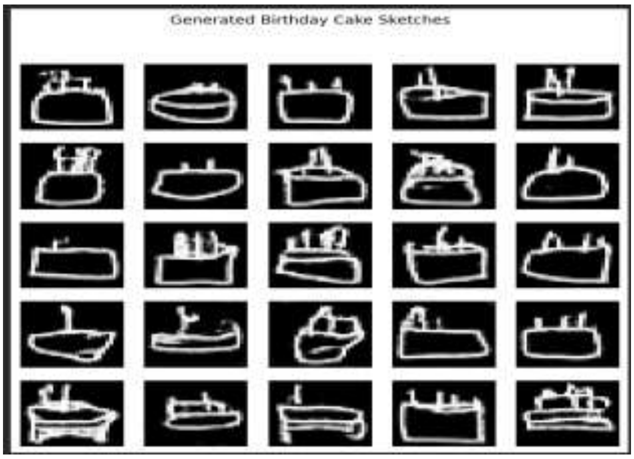


The loss behaviour showed the competition between the generator and discriminator. During early training stages, the discriminator was able to easily identify generated sketches. As training progressed, the generator improved and produced samples with characteristics closer to the original dataset.

5.4 Evaluation of Generated Sketches

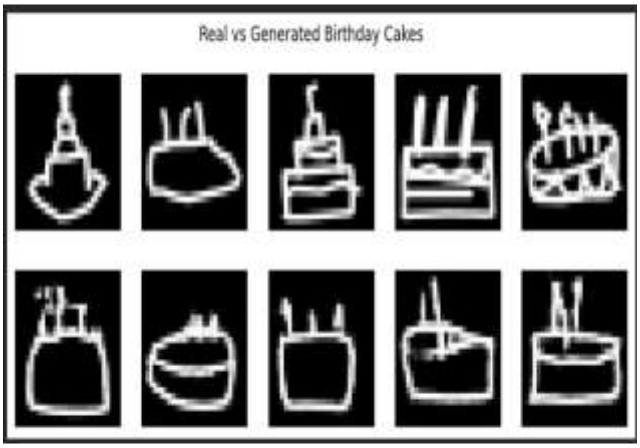
The generator was given random latent vectors to generate new birthday cake sketches after training.

Figure 13: Generated birthday cake sketches.



Basic cake shapes, vertical candle structures, and outline-based patterns were among the key elements of birthday cake drawings that were captured in the produced samples. Nevertheless, some of the produced sketches lacked the diversity found in the actual dataset and had incomplete shapes.

Figure 14: Real and produced birthday cake sketches are compared.



The comparison shows that the model was able to grasp the general idea of a birthday cake. However, because QuickDraw photos cover a wide range of human sketching styles, the produced sketches were less detailed than actual examples.

5.5 Discussion and Improvements

The findings show that DCGANs may produce new images that are similar to the original category by learning visual patterns from basic sketch datasets. Although several shortcomings were noted, the model was successful in capturing the general structure of birthday cakes.

Reduced diversity among generated samples was one drawback. It's possible that the generator did not fully capture all of the diversity in the dataset because some of the results seemed similar to one another. This issue, which is sometimes called "mode collapse" in GAN training, occurs when the generator generates a restricted range of outputs.

Potential enhancements consist of:

- To improve feature learning, train for more epochs.
- expanding the generator and discriminator networks' size and complexity.
- controlling generated features with a Conditional GAN (cGAN).
- using more sophisticated generative models, including diffusion models.

Other QuickDraw categories like cars, houses, and cats might easily be added to the model. It would be possible to further analyze how well GANs generalize across creative areas by introducing varying levels of drawing complexity through different categories.

6. Overall Discussion and Future Improvements

The use of Generative Adversarial Networks (GANs) to provide synthetic data for various purposes was examined in this project. The trials shown that the choice of network design and the type of data have an impact on GAN performance.

The GAN was able to learn basic two-dimensional distributions in Part 1. A simple fully linked GAN can learn basic patterns, as demonstrated by the sine-wave experiment. The model had to learn several sets of data points, which made the combination of Gaussian experiment more difficult. The findings demonstrated that the GAN was able to learn more complex distributions when the network complexity was increased. Because convolutional layers can learn crucial picture attributes including edges, forms, and textures, DCGAN architectures were better suited for image generating tasks like OCTMNIST retinal images and QuickDraw sketches. Although some details and changes were not accurately replicated, the generated images demonstrated that the models were able to understand the fundamental structure of the datasets.

The GAN effectively produced artificial network traffic samples for the cybersecurity application. The low mean feature difference (0.0448) suggested that the created samples had comparable statistical characteristics to the original dataset, and the PCA visualization revealed similarities between the generated and genuine data.GAN models do have several drawbacks, though. The generator may only produce a small range of samples, thus training may be erratic. Using larger datasets, training for more epochs, and utilizing sophisticated GAN techniques like Conditional GANs or Wasserstein GANs can all improve the quality of generated data.Future research might use conditional GANs to generate particular classes of medical images or particular kinds of cyberattacks. Further advancements in the creation of synthetic data may be obtained by contrasting GANs with more recent generative models, such as diffusion models.

7. Conclusion

This project showed how GANs are used in a variety of fields, including as cybersecurity, medical imaging, synthetic data production, and creative AI.The trials demonstrated that distinct GAN structures are needed for various datasets. While DCGANs performed better for image-based tasks because to their ability to learn spatial information, fully connected GANs were useful for numerical data.Synthetic samples that mirrored the features of the original datasets could be produced by the models. GANs may effectively learn complex data distributions, according to evaluation using quantitative metrics, loss curves, and visual comparisons.Even though GANs are strong generative models, issues like maintaining output diversity and training stability still exist. The quality and dependability of produced synthetic data may be significantly enhanced in the future by utilizing more sophisticated architectures and bigger datasets.

References

- Tu, Z. (2007). Learning generative models via discriminative approaches. In Computer Vision and Pattern Recognition, 2007. CVPR’07. IEEE Conference on, pages 1–8. IEEE.
- Radford, A., Metz, L. and Chintala, S., 2015. Unsupervised representation learning with deep convolutional generative adversarial networks. *arXiv preprint arXiv:1511.06434*.
- Yang, J., Shi, R., Wei, D. *et al.* MedMNIST v2 - A large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Sci Data* **10**, 41 (2023). https://doi.org/10.1038/s41597-022-01721-8.